

A healthcare cost proxy from UKHLS alone

What is buildable now, how to validate it, and when linkage becomes worth it

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Companion to `health_measures_note.pdf` §8. That section established that expected cost, not a binary event, is the right outcome for the framework’s convexity question, and that UKHLS carries the physical quantities cost is bought with. This is the plan for turning those quantities into a defensible cost proxy without linkage.

1 What we are building

A per-person-wave estimate of annual NHS secondary-care cost, in a fixed price year, for waves 7–15. It will be wrong in level and, we hope, roughly right in *shape* — which is all the framework needs, since every question we are asking is about how cost varies with health, not about its absolute size. The one-line target:

$$\hat{C}_{it} = \underbrace{c_{\text{spell}}(k) + c_{\text{bed}} \cdot (\text{nights}_{it} - \bar{L})^+}_{\text{in-patient}} + \underbrace{c_{\text{op}} \cdot \hat{n}_{it}^{\text{op}}}_{\text{out-patient}}$$

with k the attributed condition where we have one, and everything indexed to a single price year.

2 Ingredients we already hold

Quantity	Source	Status
Any admission	<code>hosp</code> , waves 7–15	complete for respondents
Nights	<code>hospd</code>	all admitted; median 3, mean 7.5, max 300
Maternity flag	<code>hospch</code>	12.7% of admissions, mean age 31.7
Nights by new condition	<code>hospdci</code> , waves 2–15	21% of admissions
Nights by prior condition	<code>hospdcp_i</code> , waves 11–15	11% of admissions
either		29% of admissions, 65% of nights
Out-patient band	<code>h12gp</code> , <code>h12hop</code>	banded 0/1–2/3–5/6–10/>10

Unit costs come from the NHS National Cost Collection, which publishes national average unit costs annually and free, at both aggregate and HRG level. Indicative recent figures: non-elective in-patient episode $\approx$ £1,590, elective $\approx$ £3,684, day case $\approx$ £738, excess bed day $\approx$ £313, out-patient attendance $\approx$ £120.

3 Four stages

Stage 1 — a flat-rate baseline (half a day)

Cost each admission at the non-elective episode rate plus excess bed days beyond the average inclusive length of stay, and each out-patient attendance at the flat rate, with band midpoints 0, 1.5, 4, 8, 15. Exclude maternity. This is deliberately crude and exists to give every later stage something to be compared against.

The one real modelling decision here. UKHLS gives total nights but *not the number of spells*, so a person with 20 nights might have had one long admission or four short ones, which differ in cost by a factor of about three. Stage 1 assumes one spell. Stage 4 tests how much that matters; if the answer is “a lot”, it becomes the main argument for linkage, because HES would resolve it exactly.

Stage 2 — condition weighting (one to two days)

For the 65% of nights carrying an attribution, replace the flat rate with a condition-specific one. The mapping is UKHLS condition code → NCC HRG chapter → average cost, at chapter level rather than individual HRG, since the survey identifies a condition and not a procedure. The attributed nights concentrate in stroke, COPD, asthma, osteoarthritis, MI, congestive heart failure and chronic kidney disease, which span a wide cost range, so this is where the proxy should improve most.

Unattributed nights take the case-mix-weighted average of the attributed ones *within the same health decile*, rather than a global average — otherwise the imputation flattens exactly the gradient we are trying to measure.

Stage 3 — validation against published profiles (one day)

This is the stage that decides whether the proxy is usable, and it is the reason to prefer building it here rather than assuming a shape.

1. **The age profile.** The CHE reference tables give average annual NHS spend by age, sex and deprivation, and the IFS has published hospital spending profiles from NHS administrative records. Our proxy should reproduce the shape — flat to about 50, rising steeply after 70 — without having been fitted to it. A proxy that misses the age profile cannot be trusted on the health profile.
2. **The concentration.** Published figures put the top decile of spenders at more than half of expenditure. Ours should land near that. Too little concentration means the intensive margin is being under-weighted; too much means outliers are running the estimate.
3. **The level.** Per-capita secondary care spend implied by the proxy against national totals. We expect to *undershoot*, because UKHLS misses A&E, and self-reported nights are known to be under-recalled; the size of the shortfall is itself informative.

Stage 4 — use it, and test what it rests on (one to two days)

Redo §8’s convexity analysis with $\mathcal{L}$ on the vertical axis, and §11’s outcome prediction with cost as the outcome. Then the sensitivity analysis that matters:

- **Spells.** Re-run assuming one spell, two spells, and nights/ $\bar{L}$ spells. If the convexity conclusion is stable across these, the spell ambiguity is not binding and linkage buys less than it looks.
- **Outlier handling.** The 300-night maximum will dominate any mean. Winsorise at the 99th percentile and re-run.
- **Condition weights.** Flat rate against condition-weighted: does the shape change, or only the level?
- **Maternity.** Already known to move the gradient from $31.9\times$ to $52.7\times$; carry both through.

4 What this cannot do

Primary care, prescribing, A&E and community services are absent entirely, and they are not a small share of NHS spend. Out-patient use is a four-band count, so its intensive margin is crude. Elective and emergency admissions are indistinguishable, though they differ in cost by more than a factor of two. And every quantity is self-reported over a twelve-month recall window, against wave spacing that is not exactly twelve months.

The proxy is therefore a *secondary-care bed-day-weighted cost index*, and should be named that way in anything we write rather than called a cost.

5 The decision this sets up

If Stage 3 shows the proxy reproducing the published age profile and concentration, and Stage 4 shows the convexity conclusion stable across the spell and outlier assumptions, then UKHLS alone supports the framework’s empirical claim and linkage is a refinement rather than a precondition.

If instead the shape moves with the spell assumption, or the proxy misses the published profiles, then the case for UK LLC linkage becomes concrete and specific — and we would know exactly which quantity we needed it for, which is a much stronger application than asking for HES in general. The binding risk there remains sample size: roughly 8,000 consented participants against the 84,679 in our current sample, with an extension to all participants stated but needing verification.

Effort

Three to five days end to end, entirely on data already on disk, with no new permissions. Stages 1 and 3 are the ones worth doing even if we stop there: a validated flat-rate index would already replace the physical-units analysis of §8 with something denominated in pounds.